EEG Optimizer

Instruction Manual - EEG analysis for the P300 effect and the alpha/beta ratio

1. What this tool does

The EEG Optimizer reads one EEG recording and answers two questions. In **P300 mode** it measures whether the brain produced a P300 oddball response, that is, whether target flashes produced a larger positive wave than non-target flashes. In **Alpha/Beta mode** it measures the ratio of alpha power to beta power, which is commonly used as an index of relaxation versus alertness.

Whichever mode you choose, three numbers are reported: **signal-to-noise ratio**, **usability** and **effect size**. Section 5 explains all three.

> Everything runs on the CPU. No graphics card is required, and no file is ever sent over the internet - the interface is a local web page served by your own computer.

2. Installing

You need **Python 3.10 or newer**. During the Python installation, tick *Add python.exe to PATH*.

1. Double-click **install.bat** in this folder.
2. Wait until it prints *Done*, then close the window.

If you prefer the command line, open a terminal in this folder and run the single command below. It installs every library listed in **requirements.txt**.

```
pip install -r requirements.txt
```

3. Starting the tool

1. Double-click **run.bat**. A black window opens and stays open - that window is the program, so do not close it while you are working.
2. Your browser opens at **http://127.0.0.1:5000**. If it does not, type that address into the browser yourself.
3. To stop the tool, click the black window and press **Ctrl + C**, or simply close it.

The **English / ■■** switch is in the top right corner of every page, including the results page.

4. Choosing the right file

Three formats are accepted. The table shows what each one supports.

Format	What it is	P300	Alpha/Beta
.edf	EDF or EDF+ containing real EEG electrodes	Yes, if it has markers	Yes
.csv	EMOTIV Pro export, named *_Original.csv or *_Cleansed.csv	Yes	Yes
.xlsx	Epoched P300 workbook with a Training_Data sheet	Yes	Not yet - see section 8

> **Important:** EMOTIV exports two files with the .edf extension. One holds the EEG; the other holds only head-motion data (quaternion, accelerometer and magnetometer at 64 Hz) and contains no brain signal at all. If you load the motion file, the tool refuses it and tells you so, rather than producing a meaningless number. When in doubt, use the matching **_Original.csv**, which always contains the EEG.

For P300 mode the file must also say which flash was a target. The tool finds this automatically from the **MarkerValueInt** column, where **51** marks a target flash and **50** marks a non-target flash. An epoched workbook uses its **label** column instead (1 = target, 0 = non-target). If your EEG file has no markers at all, upload the matching ***_markers.csv** in the optional marker box.

5. Reading the results

5.1 Signal-to-noise ratio (SNR)

In P300 mode, SNR is the height of the target-minus-non-target peak inside the 250-600 ms window, divided by the size of the random fluctuation in the 200 ms before the flash. It answers: *is the response bigger than the background noise?* In Alpha/Beta mode, SNR is how far the alpha peak rises above the neighbouring 4-7 Hz and 15-18 Hz background, which answers the same question for the alpha rhythm.

Value	Meaning
3.0 or above	Clean. The effect stands clearly above the noise.
1.5 to 3.0	Usable but modest. Interpret alongside the effect size.
Below 1.5	The effect is close to the noise floor. Treat with caution.

5.2 Usability (0-100)

Usability answers a different question from SNR: *can this recording be trusted at all?* It is a weighted blend of four checks, shown as a dial with a coloured badge, and broken down in a table on the results page.

Check	Weight	What it measures
Artifact control	35	Share of epochs or segments discarded for excessive amplitude. 20 percent rejection scores half marks; 40 percent scores zero.
Trial count / Recording length	25	P300 mode wants at least 30 clean target epochs; Alpha/Beta mode wants at least 60 seconds.
Signal clarity	25	The SNR from section 5.1, scaled against the target of 3.0.
Channel integrity	15	Share of channels carrying live, non-flat signal. In Alpha/Beta mode this also penalises 50 Hz mains interference.

Score	Badge	What to do
70 - 100	PASS	Use the recording.
45 - 69	REVIEW	Look at the plots before drawing conclusions.
Below 45	FAIL	Re-record if you can. Check electrode contact and the saline.

> A high usability score does not prove a P300 exists. It only says the recording is clean enough for the question to be asked. Usability and effect size must be read together.

5.3 Effect size (Cohen's d)

In P300 mode, Cohen's d measures how far apart the target and non-target epochs sit, in units of their own spread. It is the number that says how strong the P300 is.

Cohen's d	Verdict	Meaning
0.30 or above	STRONG	Clear separation between target and non-target.
0.15 - 0.29	MODERATE	A real but modest oddball response.
Above 0, below 0.15	WEAK	Close to the noise floor.
0 or negative	ABSENT	No positive target-minus-non-target deflection was found.

In Alpha/Beta mode, Cohen's d compares alpha power against beta power across the recording's time segments. A **positive** value means alpha consistently leads; a **negative** value means beta leads. The further from zero, the more consistent that dominance is.

5.4 Extra numbers on the results page

Field	Meaning
Clearest channel	The centro-parietal electrode with the largest positive P300 peak. Pz and Cz are the classic sites.
Peak latency	When the peak occurred. A genuine P300 usually falls between 250 and 500 ms after the flash.
Alpha / Beta ratio	Above 2.0 relaxed, 1.0 to 2.0 balanced, 0.5 to 1.0 alert, below 0.5 highly alert.
Mains noise (50 Hz)	How far the 50 Hz peak rises above its neighbours. 1.0 means no interference; above 3.0 suggests a nearby power cable.

6. Saving your results

Two buttons sit at the bottom of the results page. **Download PDF report** produces a formatted report containing every number and every figure, in the language the interface is currently set to - switch to ■■ first if you want a Chinese report. **Download plots** gives you a zip of the PNG figures for pasting into a thesis or slide deck.

7. If something goes wrong

Message	What to do
contains no EEG. It is an EMOTIV motion export	You picked the motion file. Use the FLEX2-named .edf or the _Original.csv for that session.
carries no target/non-target flash markers	The file has no MarkerValueInt codes 51/50. This is normal for a baseline recording - use Alpha/Beta mode, or upload the matching *_markers.csv.
None of the centro-parietal channels are present	The file has no Pz, Cz, Fz, P3, P4, CP1 or CP2. P300 cannot be measured without at least one of them.
Too few target or non-target epochs	Artifact rejection removed too much. The recording is probably very noisy; check electrode contact.
Python was not found	Install Python 3.10 or newer and tick *Add python.exe to PATH*.
The browser page will not open	Make sure the black window is still running, then type http://127.0.0.1:5000 into the browser.

8. Future work

Alpha/beta ratio from epoched Excel workbooks. Choosing Alpha/Beta mode with an .xlsx file shows a *Coming Soon* page rather than a number. The reason is a property of the files themselves: when those workbooks were created they were band-pass filtered to 0.1-15 Hz, and the beta band runs from 13 to 30

Hz. Almost all beta power has therefore already been removed, so any ratio computed from them would be badly inflated and misleading. Supporting this properly requires re-exporting the epochs with a wider filter. Until then, use the matching .edf or _Original.csv, which keep the full frequency range.

9. Technical notes

Setting	Value
P300 epoch window	-200 ms .. +800 ms
Baseline correction	-200 ms .. 0 ms
P300 search window	250 ms .. 600 ms
P300 band-pass	0.1 - 15 Hz, 4th order Butterworth, zero phase
Alpha/Beta band-pass	1 - 45 Hz, 4th order Butterworth, zero phase
Frequency bands	delta 1-4, theta 4-8, alpha 8-13, beta 13-30, gamma 30-45 Hz
Spectral estimate	Welch, 4 s windows, 50 percent overlap
Artifact rejection	Worst-channel peak-to-peak above median + 4 x MAD
P300 report channels	Pz, Cz, Fz, P3, P4, CP1, CP2
Alpha/Beta default channels	O1, O2, Oz, Pz
Compute device	CPU only

Artifact rejection scales to each recording's own amplitude distribution rather than using a fixed microvolt threshold, because raw EMOTIV Flex values are not calibrated microvolts. Amplitudes should therefore be compared within a recording, not between recordings made on different days or with different saline.

> These measurements describe the recorded signal. They are a quality gate before modelling, not a claim about how accurately a classifier will decode intent.

EEG 脑电分析器

使用说明书 - P300 效应与 α / β 波比值脑电分析

1. 本工具的用途

EEG 脑电分析器读取一份脑电记录并回答两个问题。在 P300 模式 下，它检测大脑是否产生了 P300 奇异刺激反应，即目标闪烁是否比非目标闪烁产生了更大的正向波形。在 α / β 模式 下，它计算 α 波能量与 β 波能量的比值，该比值常用于衡量放松与警觉的程度。

无论选择哪种模式，都会给出三个指标：信噪比、可用性 与 效应量。第 5 节将逐一说明。

> 所有计算均在 CPU 上完成，无需显卡；任何文件都不会上传到网络 —— 界面只是本机提供的一个网页。

2. 安装

需要 Python 3.10 或更新版本。安装 Python 时请勾选 Add python.exe to PATH。

1. 双击本文件夹中的 install.bat。
2. 等待窗口显示 Done 后关闭。

如果习惯使用命令行，可在本文件夹中打开终端并执行下面这一条命令，它会安装 requirements.txt 中列出的全部库。

```
pip install -r requirements.txt
```

3. 启动工具

1. 双击 run.bat。会出现一个黑色窗口并保持打开 —— 该窗口就是程序本身，使用期间请勿关闭。
2. 浏览器会自动打开 http://127.0.0.1:5000。若未自动打开，请手动输入该地址。
3. 结束时，点击黑色窗口并按 Ctrl + C，或直接关闭该窗口。

每个页面右上角都有 English / 中文 切换按钮，结果页面也不例外。

4. 选择正确的文件

支持三种格式，下表说明各自的适用范围。

格式	文件说明	P300	α / β
.edf	包含真实脑电电极的 EDF 或 EDF+	有标记时可用	可用
.csv	EMOTIV Pro 导出文件，文件名为 *_Original.csv 或 *_Cleansed.csv	可用	可用
.xlsx	含 Training_Data 工作表的分段 P300 工作簿	可用	暂不支持，见第 8 节

> 重要提示： EMOTIV 会导出两个 .edf 文件。其中一个是脑电数据，另一个只包含头部运动数据（四元数、加速度计与磁力计，64 Hz），完全没有脑信号。若误选运动数据文件，本工具会明确拒绝并提示，而不会给出无意义的数值。如不确定，请使用对应的 _Original.csv，其中一定包含脑电数据。

P300 模式还需要知道哪些闪烁是目标刺激。本工具会自动从 MarkerValueInt 列读取：51 表示目标闪烁，50 表示非目标闪烁。分段工作簿则使用其 label 列（1 = 目标，0 = 非目标）。若脑电文件完全没有标记，请在可选标记框中上传对应的 *_markers.csv。

5. 解读分析结果

5.1 信噪比 (SNR)

在 P300 模式下，信噪比是 250-600 毫秒窗口内「目标减非目标」峰值的高度，除以闪烁前 200 毫秒内随机波动的大小。它回答的问题是：该反应是否大于背景噪声？在 α / β 模式下，信噪比是 α 峰高出相邻 4-7 Hz 与 15-18 Hz 背景的倍数，对 α 节律回答同样的问题。

数值	含义
3.0 及以上	干净。效应明显高于噪声。
1.5 至 3.0	可用但一般，需结合效应量判断。
低于 1.5	效应接近噪声水平，须谨慎对待。

5.2 可用性 (0-100)

可用性回答的问题与信噪比不同：这份记录整体是否可信？它由四项检查加权合成，以圆环仪表和彩色徽章显示，并在结果页面的表格中逐项列出。

检查项	权重	衡量内容
伪迹控制	35	因幅度过大而被剔除的分段比例。剔除 20% 得一半分，剔除 40% 得零分。
试次数量 / 记录时长	25	P300 模式需至少 30 个干净的目标分段；α / β 模式需至少 60 秒记录。
信号清晰度	25	第 5.1 节的信噪比，以 3.0 为满分标准换算。
通道完整性	15	有效且非平直信号的通道占比。在 α / β 模式下还会因 50 Hz 工频干扰扣分。

得分	徽章	建议
70 - 100	合格	可以使用该记录。
45 - 69	需复查	下结论前请先查看图表。
低于 45	不合格	如有条件请重新采集，并检查电极接触与生理盐水。

> 可用性高并不代表一定存在 P300，它只说明该记录足够干净、值得分析。可用性与效应量必须结合起来解读。

5.3 效应量 (Cohen's d)

在 P300 模式下，Cohen's d 以数据自身的离散程度为单位，衡量目标与非目标分段之间的分离程度，是判断 P300 强弱的核心数值。

Cohen's d	判定	含义
0.30 及以上	强	目标与非目标分离清晰。
0.15 - 0.29	中等	存在真实但较弱的奇异反应。
大于 0 且小于 0.15	弱	接近噪声水平。
小于等于 0	未检出	未发现「目标减非目标」的正向偏移。

在 α / β 模式下，Cohen's d 比较各时间段中的 α 能量与 β 能量。正值 表示 α 持续占优，负值 表示 β 占优。绝对值越大，这种优势越稳定。

5.4 结果页面的其他数值

字段	含义
最清晰通道	P300 正向峰值最大的中央-顶叶电极，Pz 与 Cz 是经典位置。
峰值潜伏期	峰值出现的时间。真实的 P300 通常在闪烁后 250 至 500 毫秒之间。
α / β 比值	大于 2.0 为放松，1.0 至 2.0 为平衡，0.5 至 1.0 为警觉，小于 0.5 为高度警觉。
工频干扰 (50 Hz)	50 Hz 峰高出相邻频段的倍数。1.0 表示无干扰，大于 3.0 说明附近可能有电源线干扰。

6. 保存分析结果

结果页面底部有两个按钮。下载 PDF 报告 会生成包含全部数值与图表的正式报告，语言与当前界面一致 —— 若需中文报告，请先切换到 中文。下载图表 会提供 PNG 图片的压缩包，方便插入论文或幻灯片。

7. 常见问题处理

提示信息	处理方法
不包含脑电数据，是 EMOTIV 运动数据文件	选错了运动数据文件。请改用带 FLEX2 名称的 .edf 或该次记录的 _Original.csv。
没有目标/非目标闪烁标记	文件中没有 MarkerValueInt 的 51/50 代码。基线记录属于正常情况 —— 请改用 α / β 模式，或上传对应的 *_markers.csv。
缺少中央-顶叶通道	文件中没有 Pz、Cz、Fz、P3、P4、CP1 或 CP2。缺少这些通道则无法测量 P300。
目标或非目标分段过少	伪迹剔除过多，记录可能噪声很大，请检查电极接触。
找不到 Python	请安装 Python 3.10 或更新版本，并勾选 Add python.exe to PATH。
浏览器页面无法打开	确认黑色窗口仍在运行，然后在浏览器中输入 http://127.0.0.1:5000。

8. 后续工作

基于分段 Excel 工作簿的 α / β 比值。在 .xlsx 文件上选择 α / β 模式时，会显示 敬请期待 页面而非数值。原因在于文件本身的特性：这些工作簿在生成时已被带通滤波至 0.1-15 Hz，而 β 频段为 13 至 30 Hz。因此绝大部分 β 能量已被滤除，由此算出的比值会被严重放大并产生误导。要正确支持该功能，需以更宽的滤波范围重新导出分段数据。在此之前，请使用保留完整频率范围的 .edf 或 _Original.csv。

9. 技术说明

参数	取值
P300 分段窗口	-200 ms .. +800 ms
基线校正	-200 ms .. 0 ms
P300 搜索窗口	250 ms .. 600 ms
P300 带通滤波	0.1 - 15 Hz，4 阶巴特沃斯，零相位
α / β 带通滤波	1 - 45 Hz，4 阶巴特沃斯，零相位
频段划分	δ 1-4、θ 4-8、α 8-13、β 13-30、γ 30-45 Hz
谱估计方法	Welch 法，4 秒窗口，50% 重叠
伪迹剔除	最差通道峰峰值超过 中位数 + 4 x MAD
P300 报告通道	Pz, Cz, Fz, P3, P4, CP1, CP2
α / β 默认通道	O1, O2, Oz, Pz
计算设备	仅使用 CPU

伪迹剔除依据每份记录自身的幅度分布进行缩放，而非使用固定的微伏阈值，因为 EMOTIV Flex 的原始数值并非经过校准的微伏值。因此幅度只应在同一份记录内部比较，不宜跨日期或跨不同生理盐水的记录进行比较。

> 上述指标描述的是所记录的信号本身，属于建模前的质量把关，并不代表分类器解码意图的准确率。